MECHATRONICS ENGINEERING PROGRAM

Mission

The mission of the Mechatronics Engineering program at RHU is to provide a solid education that enables students to start a professional career and/or continue graduate studies in mechatronics engineering. The program seeks to instill in students a sense of and understanding of moral and professional responsibilities, the ability to learn more, and the leadership required in their calling, primarily in Lebanon and the Middle East.

Objectives

In a few years after graduation, Mechatronics Engineering program graduates will:

1. Apply their multidisciplinary knowledge in Mechatronics engineering to meet employers' expectations.
2. Possess self-learning motivation and qualifications to pursue advanced studies.
3. Demonstrate professional skills and embrace leadership roles in their profession and/or communities.

Program Outcomes

Upon successful completion of the BS/BE program in Mechatronics Engineering, graduates will have:

1. Ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. Ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. Ability to communicate effectively with a range of audiences
4. Ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. Ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. Ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Accreditation

The Bachelor of Engineering program in Mechatronics Engineering is accredited by the Engineering Accreditation Commission of ABET (https://www.abet.org) under the commission's General Criteria, with no applicable program criteria.

Career Opportunities

Mechatronics engineers attain a broad spectrum of skills that are central to the practice of almost any professional field. Mechatronics engineers can find career opportunities in industries and enterprises such as food processing, building and construction, medical and pharmaceutical industries, aerospace, automotive, chemical, oil and gas, computer and electronics, renewable energy, and far more. Most importantly, you will be prepared to imagine something that never was and drive it forward to make it a reality! There is so much to be gained in ensuring the future of humanity, and mechatronics engineers can tap into these possibilities through the spirit of innovation and entrepreneurship.

Program Overview

At RHU, every experience the student has represents a block in building a competent, confident, purposeful, problem-solving, competitive, responsible, and conscientious individual. Mechatronics engineering, being a holistic approach to designing and developing multifaceted smart systems, has remarkable strength. Student transformation is accomplished through a curriculum that orbits the edges of knowledge, facilities equipped with the latest technology and conforming to the highest standards, faculty members who are passionate and committed to students' academic success and personal growth, and an environment that inspires learning and drives discovery.

The Bachelor of Engineering (BE) program in Mechatronics Engineering comprises 152 credit hours across eight regular semesters and three Summer Semesters. The Bachelor of Science (BS) program comprises 118 credit hours across six regular semesters and three Summer Semesters. In addition to completing the credit requirements, a BS or a BE degree is conferred upon a student if the earned cumulative GPA, major GPA, and the Summative Learning Project grades are all 70 or above.

The required credit hours span three categories: University requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of credit hours for the BS and BE programs is shown in the following tables, which reveal the apparent multidisciplinary nature of Mechatronics. This demonstrates the value of a degree in Mechatronics as a starting point for several higher degree paths.

BS in Mechatronics Engineering (118 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	0	0	21	6	27	23

College Requirement	0	0	24	0	24	20
Program Requirement	28	0	39	0	67	57
Credits	28	0	84	6	118	100

BE in Mechatronics Engineering (152 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	0	0	21	9	30	20
College Requirement	0	0	27	0	27	18
Program Requirement	41	15	39	0	95	62
Credits	41	15	87	9	152	100

I. University Requirements (General Education)

The list of the university's required courses and their descriptions is presented in the introductory pages of the College of Engineering section in this catalog.

II. College Requirements

The list of required courses and descriptions is presented in the introductory pages of the College of Engineering section in this catalog.

III. Program Requirements

The program requirements for a Bachelor of Engineering degree in Mechatronics Engineering encompass 95 credits (67 credits for the BS degree), distributed as follows: 80 credits (67 credits for the BS degree) of mandatory courses and 15 credits of electives.

A. Mandatory Requirements

The mandatory core courses in Mechatronics Engineering (major and non-major) are listed in the table below.

Course #	Title	Credits	Prerequisites
CCEE 221	Logic Design	3	
CCEE 221L	Logic Design Lab	1	CCEE 221
CIVE 211	Statics	3	
ELEC 320	Electronics	3	ELEC 210
ELEC 320L	Electronics Lab	1	Co-req.: ELEC 320
ELEC 331	Electric Machines	3	ELEC 210
MECA 340	Dynamic Systems Modeling	3	MATH 314

MECA 341	Measurements and Instrumentation	3	ELEC 320
MECA 341L	Measurements and Instrumentation Lab	1	Co-req.: MECA 341
MECA 400*	Summative Learning (BS) Project	3	ENGL 217
MECA 440	Control Systems Design	3	MECA 340
MECA 440L	Control Systems Design Lab	1	MECA 440
MECA 441	Sensors and Actuators	3	MECA 341
MECA 442	Microcontrollers for Mechatronics	3(2+2)	CCEE 221 and MECH 200
MECA 443	Mechatronic System Design	3	MECA 440, 441, and 442
MECA 443L	Mechatronic System Design Lab	1	Co-req.: MECA 443
MECA 444L	Mechatronic Systems Programming	1	CCEE 214 and MECH 200
MECA 445	Advanced Programming for AI and Robotics	3	MECA 444L
MECA 444	Mechatronic Systems Programming	3	CCEE 214 and MECH 200
MECA 499	Co-op Work Experience	3	ENGL 217 Senior Standing
MECA 541	CAD/CAM/CAE	3	MECH 224 and MECH 421
MECA 542	Industrial and Manufacturing Control	3	ELEC 320
MECA 544	Robotics	3	MECH 321
MECA 595A	BE Summative Learning Project I	1	ENGL 217 Senior Standing
MECA 595B	BE Summative Learning Project II	3	MECA 595A
MECA 4xx/5xx	Technical Electives	15	Per course requirement
MECH 200	Introduction to Mechanical Engineering	3	
MECH 220	Dynamics	3	CIVE 211
MECH 224	Mechanical Engineering Graphics & CAD	3	
MECH 320	Mechanics of Materials	3	CIVE 211
MECH 320L	Mechanics of Materials Lab	1	Co-req.: MECH 320
MECH 321	Mechanisms and Dynamics of Machinery	3	MECH 200 and MECH 220
MECH 333	Thermal Fluid Sciences	3	MATH 216
MECH 421	Mechanical Design I	3	MECH 320
MECH 422	Manufacturing Processes	3	MECH 320

* Required for BS-bound students. BE students take a Technical Elective instead.

B. Technical Electives

As part of the Bachelor of Engineering in Mechatronics Engineering program, the student is required to take 15 credit hours of technical electives. These courses allow students to focus on a specific area to develop in-depth knowledge and understanding. The student can also mix and match elective courses from different areas to gain a broader exposure to interrelated Engineering disciplines. The student should, in cooperation with the academic advisor, select a list of electives that best meet their needs and aspirations. The listed technical elective courses and other courses from other engineering programs, in addition to the required program courses, are designed to allow the student to develop in-depth knowledge and understanding in the following areas:

- Mechanisms and Robotics
- Automation

The student is highly recommended to take elective courses after completing the mandatory requirements in the related area. Students may select technical electives from other programs, subject to the approval of the academic advisor and the corresponding course instructor.

Course #	Title	Credits	Prerequisites
Mechanisms and Robotics			
MECA 485	Autotronics	3	MECA 440
MECA 498	Special Topics in Mechatronics Engineering	3	
MECA 535	Expert Systems and Applied Intelligence	3	MECA 444L
MECA 536	Applied Deep Learning for Intelligent Systems	3	MECA 444L
MECA 543	Fluid Power Systems	3	MECH 333 or MECH 335
MECA 581	Systems Engineering	3	
MECH 522	Mechanism Synthesis Classical Kinematics to AI-Driven Optimization	3	MECH 321
Automation			
MECA 484	Civionics	3	
MECA 485	Autotronics	3	MECA 440
MECA 498	Special Topics in Mechatronics Engineering	3	
MECA 515	Modern Control Systems	3	MECA 440
MECA 545	Integrated Automation Lab	1	Co-req: MECA 542
MECA 546	Smart Energy Management Systems	3	MECA 442
MECA 586	Mechatronics Drives	3	ELEC 320

C. Summative Learning Project

Students must complete a 3-credit-hour course for the BS (4 credits for the BE; take 1 credit in the first regular semester and 3 credits in the following regular semester) in which they work, preferably in groups, on a problem of concern to industry or the community at large, or to develop a promising idea. The SLP could be an extension of

the projects students worked on in the Engineering for the Community course, bringing them to a more useful outcome.

D. Co-op Experience

Students must complete 8 weeks of practical training in a field related to their interests. The Co-op experience is usually fulfilled during the summer semester of the third year in the program. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member follows up on a student's progress throughout the Co-op period by conducting field visits and ensuring that the student's performance is aligned with their aspirations and the employer's needs. Students are required to submit a formal report and a poster and to give a formal presentation about their Co-op experience. While ENGL 217 and Senior Standing are the essential prerequisites for co-op training, the program may stipulate additional conditions.

Study Plan

The following study plan summarizes the courses and credit distribution for the Bachelor of Science (BS) and Bachelor of Engineering (BE) in MECA Engineering. The following study plan serves as a roadmap for a smooth progression toward graduation.

Course #	Title	Credits	Prerequisites
Year 1, Fall Semester (15 Credits)			
CCEE 214	Introduction to Programming	3	
CIVE 211	Statics	3	
ENGL 210	English Composition and Rhetoric	3	Placement
MATH 215	Calculus III	3	
PHYS 210	Fundamentals of Physics and Lab	3	
Year 1, Spring Semester (15 Credits)			
ELEC 210	Electric Circuits	3	PHYS 191 or Equivalent
MATH 216	Calculus IV	3	MATH 215
MECH 200	Introduction to Mechanical Engineering	3	
MECH 220	Dynamics	3	CIVE 211
	Humanities/Fine Arts Elective I	3	
Year 1, Summer Semester (9 Credits)			
CHEM 211	Environmental Chemistry and Lab	3	
MECH 224	Mechanical Engineering Graphics & CAD	3	
MATH 311	Linear Algebra with Applications	3	
Year 2, Fall Semester (17 Credits)			
CCEE 221	Logic Design	3	
ELEC 320	Electronics	3	ELEC 210
ELEC 320L	Electronics Lab	1	Co-req: ELEC 320
MATH 314	Ordinary Differential Equations	3	MATH 216
MECH 320	Mechanics of Materials	3	CIVE 211

MECH 320L	Mechanics of Materials Lab	1	Co-req.: MECH 320
MECH 321	Mechanisms and Dynamics of Machinery	3	MECH 200 and MECH 220
Year 2, Spring Semester (17 Credits)			
CCEE 221L	Logic Design Lab	1	CCEE 221
ENGL 217	Professional English Communication	3	ENGL 210
MATH 351	Probability and Statistics	3	MATH 215
MECA 340	Dynamic Systems Modeling	3	MATH 314
MECA 341	Measurements and Instrumentation	3	ELEC 320
MECA 341L	Measurements and Instrumentation Lab	1	Co-req: MECA 341
MECH 333	Thermal Fluid Sciences	3	MATH 216
Year 2, Summer Semester (9 Credits)			
ENGR 300	Engineering Economics and Management	3	
MATH 317	Partial Differential Equations	3	MATH 314
	Social Sciences/ Cultural Elective I	3	
Year 3, Fall Semester (16 Credits)			
MECA 440	Control Systems Design	3	MECA 340
MECA 441	Sensors & Actuators	3	MECA 341
MECA 442	Microcontrollers for Mechatronics	3 (2+2)	CCEE 221 and MECH 200
MECA 444L	Mechatronic Systems Programming	1	CCEE 214 and MECH 200
MECA 541	CAD/CAM/CAE	3	MECH 224 and MECH 421
MECH 421	Mechanical Design I	3	MECH 320
Year 3, Spring Semester (17 Credits)			
ARAB 212	Arabic Language & Communication	2	
ELEC 331	Electric Machines	3	ELEC 210
MATH 421	Numerical Analysis	3	MATH 311
MECA 440L	Control Systems Design Lab	1	MECA 440
MECA 443	Mechatronic System Design	3	MECA 440, 441, and 442
MECA 443L	Mechatronic System Design Lab	1	Co-req.: MECA 443
	Community & Sustainability	1	Co-req.: ENGL 210
	Engineering Technical Elective I*	3	Per course requirements
Year 3, Summer Semester (3 Credits)			
MECA 499	Co-op Work Experience	3	ENGL 217 Senior Standing
The 118 credits above complete the requirements for a Bachelor of Science Degree in Mechatronics Engineering.			
Year 4, Fall Semester (16 Credits)			
ENGR 510	Engineering Project Management	3	ENGR 300